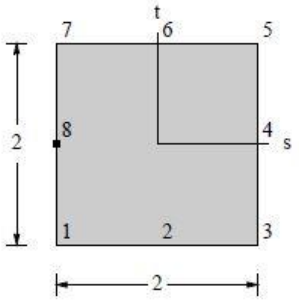
## 任意四边形插值函数与映射单元（**mapped elements**）：

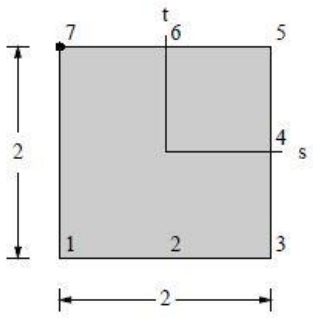
(i) Interpolation functions when all four sides are quadratic curves: 8 nodes

$$N=\begin{pmatrix} -\frac{1}{4}(-1+s)(-1+t)(1+s+t) \\ \frac{1}{2}(-1+s^2)(-1+t) \\ \frac{1}{4}(-1+t)(1-s^2+t+st) \\ -\frac{1}{2}(1+s)(-1+t^2) \\ \frac{1}{4}(1+s)(1+t)(-1+s+t) \\ -\frac{1}{2}(-1+s^2)(1+t) \\ \frac{1}{4}(-1+s)(1+s-t)(1+t) \\ \frac{1}{2}(-1+s)(-1+t^2) \end{pmatrix}$$



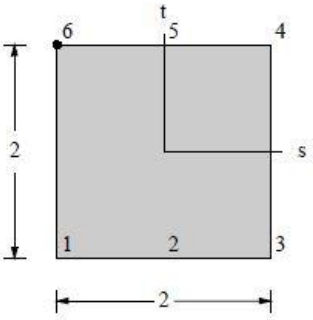

(ii) Interpolation functions when first three sides are quadratic curves: 7 nodes

$$N=\begin{pmatrix} -\frac{1}{4}(-1+s)s(-1+t) \\ \frac{1}{2}(-1+s^2)(-1+t) \\ \frac{1}{4}(-1+t)(1-s^2+t+st) \\ -\frac{1}{2}(1+s)(-1+t^2) \\ \frac{1}{4}(1+s)(1+t)(-1+s+t) \\ -\frac{1}{2}(-1+s^2)(1+t) \\ \frac{1}{4}(-1+s)s(1+t) \end{pmatrix}$$



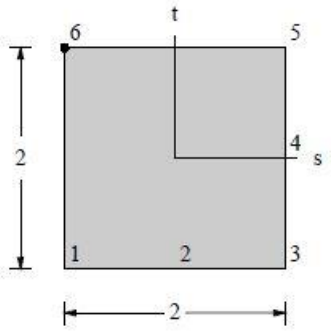

(iii) Interpolation functions when the opposite sides are quadratic curves: 6 nodes

$$N=\begin{pmatrix} \frac{1}{4}(-1+s)s(1-t) \\ -\frac{1}{2}(-1+s)(1+s)(1-t) \\ \frac{1}{4}s(1+s)(1-t) \\ \frac{1}{4}s(1+s)(1+t) \\ -\frac{1}{2}(-1+s)(1+s)(1+t) \\ \frac{1}{4}(-1+s)s(1+t) \end{pmatrix}$$



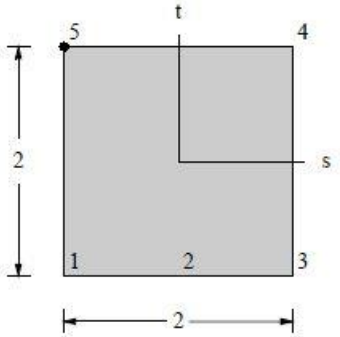

(iv) Interpolation functions when the two adjacent sides are quadratic curves: 6 nodes

$$N = \begin{pmatrix} -\frac{1}{4}(-1+s)s(-1+t) \\ \frac{1}{2}(-1+s^2)(-1+t) \\ \frac{1}{4}(-1+t)(1-s^2+t+st) \\ -\frac{1}{2}(1+s)(-1+t^2) \\ \frac{1}{4}(1+s)t(1+t) \\ -\frac{1}{4}(-1+s)(1+t) \end{pmatrix}$$




(v) Interpolation functions when the first side is a quadratic curve: 5 nodes

$$N = \begin{pmatrix} -\frac{1}{4}(-1+s)s(-1+t) \\ \frac{1}{2}(-1+s^2)(-1+t) \\ -\frac{1}{4}s(1+s)(-1+t) \\ \frac{1}{4}(1+s)(1+t) \\ -\frac{1}{4}(-1+s)(1+t) \end{pmatrix}$$




**Example 6.6** *Annular area*

Develop a mapping to a $2 \times 2$ square for the annular area shown in Figure 6.11. The inner radius is 2 units and the outer radius is 8 units. The circular arcs are defined by three points. The coordinates of all six key points are as follows.

| | x | y |
|---|---|---|
| 1 | 0 | 2 |
| 2 | $\sqrt{2}$ | $\sqrt{2}$ |
| 3 | 2 | 0 |
| 4 | 8 | 0 |
| 5 | $4\sqrt{2}$ | $4\sqrt{2}$ |
| 6 | 0 | 8 |

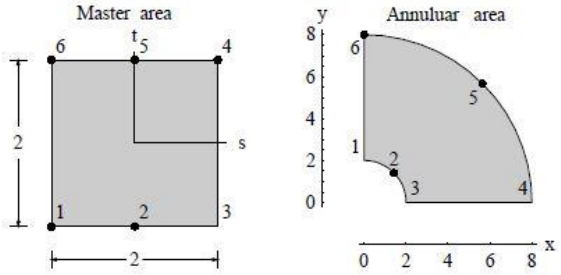



Figure 6.11. Six node master area and actual annular area

The node numbering is chosen in such a way that three nodes are placed on the first and the third side of the master element. The interpolation functions for this 6 node element are given in case (iii). Multiplying these interpolation functions by the $x$ and $y$ coordinates of the 6 key points of the annular area we get the following mapping.

$$x(s,t) = \frac{1}{4}(s-1)s(1-t)x_1 - \frac{1}{2}(s-1)(s+1)(1-t)x_2 + \frac{1}{4}s(s+1)(1-t)x_3 +$$
$$\frac{1}{4}s(s+1)(t+1)x_4 - \frac{1}{2}(s-1)(s+1)(t+1)x_5 + \frac{1}{4}(s-1)s(t+1)x_6 =$$
$$-\frac{3ts^2}{\sqrt{2}} + \frac{3ts^2}{2} - \frac{5s^2}{\sqrt{2}} + \frac{5s^2}{2} + \frac{3ts}{2} + \frac{5s}{2} + \frac{3t}{\sqrt{2}} + \frac{5}{\sqrt{2}}$$

$$y(s,t) = \frac{1}{4}(s-1)s(1-t)y_1 - \frac{1}{2}(s-1)(s+1)(1-t)y_2 + \frac{1}{4}s(s+1)(1-t)y_3 +$$
$$\frac{1}{4}s(s+1)(t+1)y_4 - \frac{1}{2}(s-1)(s+1)(t+1)y_5 + \frac{1}{4}(s-1)s(t+1)y_6 =$$
$$-\frac{3ts^2}{\sqrt{2}} + \frac{3ts^2}{2} - \frac{5s^2}{\sqrt{2}} + \frac{5s^2}{2} - \frac{3ts}{2} - \frac{5s}{2} + \frac{3t}{\sqrt{2}} + \frac{5}{\sqrt{2}}$$

The Jacobian matrix and the Jacobian of the mapping are as follows.

$$J = \begin{pmatrix} -3\sqrt{2}\,ts + 3ts - 5\sqrt{2}\,s + 5s + \frac{3t}{2} + \frac{5}{2} & -\frac{3s^2}{\sqrt{2}} + \frac{3s^2}{2} + \frac{3s}{2} + \frac{3}{\sqrt{2}} \\ -3\sqrt{2}\,ts + 3ts - 5\sqrt{2}\,s + 5s - \frac{3t}{2} - \frac{5}{2} & -\frac{3s^2}{\sqrt{2}} + \frac{3s^2}{2} - \frac{3s}{2} + \frac{3}{\sqrt{2}} \end{pmatrix}$$

$$\det J = \frac{9ts^2}{\sqrt{2}} - \frac{9ts^2}{2} + \frac{15s^2}{\sqrt{2}} - \frac{15s^2}{2} + \frac{9t}{\sqrt{2}} + \frac{15}{\sqrt{2}}$$

Setting detJ = 0 and solving for t we see that the Jacobian is zero when

$$t = -\frac{5}{3}$$

This is clearly outside of the master area and hence the mapping is good. The Figure 6.12 provides a graphically illustration of the mapping. A regular grid of points is chosen on the master area. For each pair of $s, t$ values, the corresponding $x, y$ values are computed from the mapping. All points are clearly mapped properly.

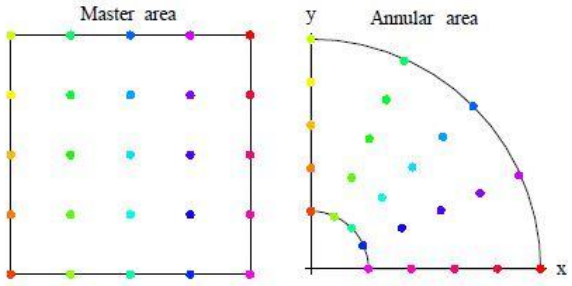



Figure 6.12. Mapping from the master area to the annular area